Internet of Things (IoT):

From constrained microcontrollers and wireless meshes to edge pipelines and semantic discovery.

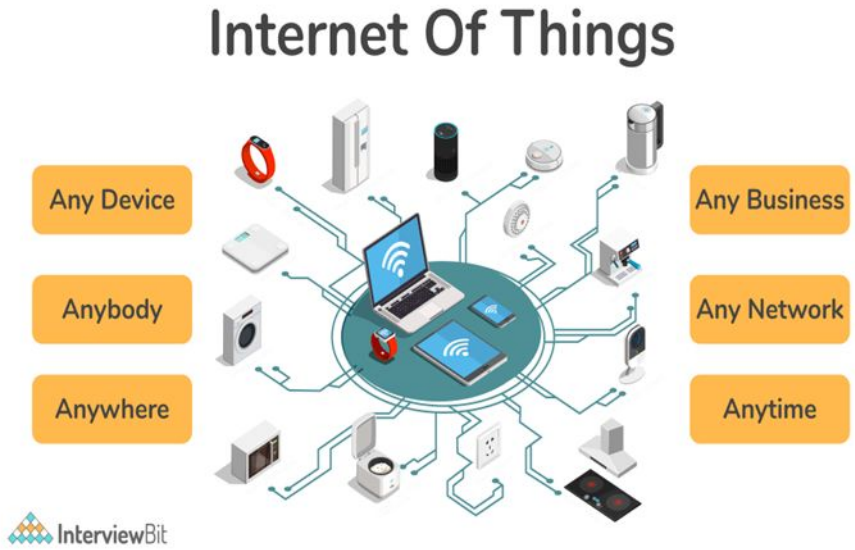
Week 2



Department of Software Engineering
Faculty of Computing
UTM

Introduction

- ✔ What is IoT?
- ✔ Anatomy of an IoT
- ✔ Sensor and Actuator
- ✔ Edge Computing vs IoT
- ✔ IoT Architecture
- ✔ Application Domains
- ✔ IoT Network Protocols
- ✔ Communication Protocols
- ✔ IoT Storage
- ✔ Semantic Intelligence
- ✔ Exercise Challenge
- ✔ Summary





What is the Internet of Things?

Cyber-Physical Convergence

Autonomous interplay between **software** algorithms and **physical** environments or objects.

Machine-to-Machine (M2M)

Automated telemetry exchange without active human intervention through **Internet**.

Embedded Intelligence

Equipping ordinary **objects** with **sensors, actuation, compute, and networking**.



"IoT or Internet of Things, refers to physical objects (or groups of objects) with sensors, processing ability, software and other technologies that connect and exchange data with other devices and systems over the Internet or other communications networks."

REAL-WORLD IMPACT

Major Application Domains

Smart Cities

Adaptive traffic signaling and dynamic utility grids.

Industrial IoT

Predictive machine maintenance via vibration analysis.

Digital Health

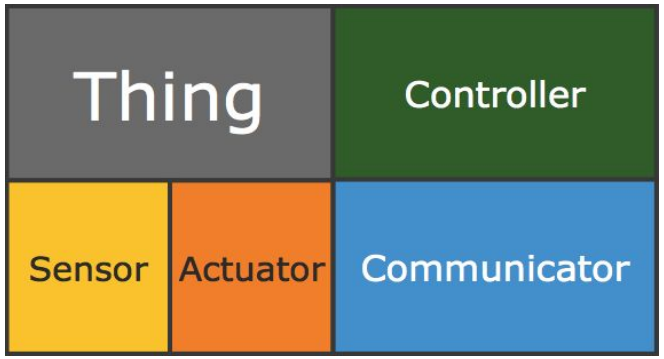
Continuous glucose telemetry and remote patient monitors.

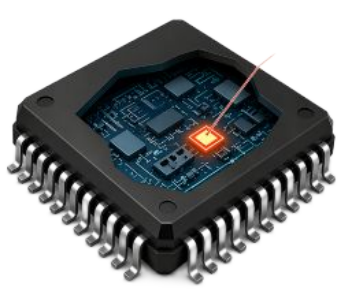
Agri-Tech

Soil hydrology profiling for autonomous irrigation.

Anatomy of an IoT

Physical object (“thing”)
+
Controller (“brain”)
+
Sensors
+
Actuators
+
Networks (Internet)







Sensors (Perception)

Translating analog physical phenomena to digital bits.

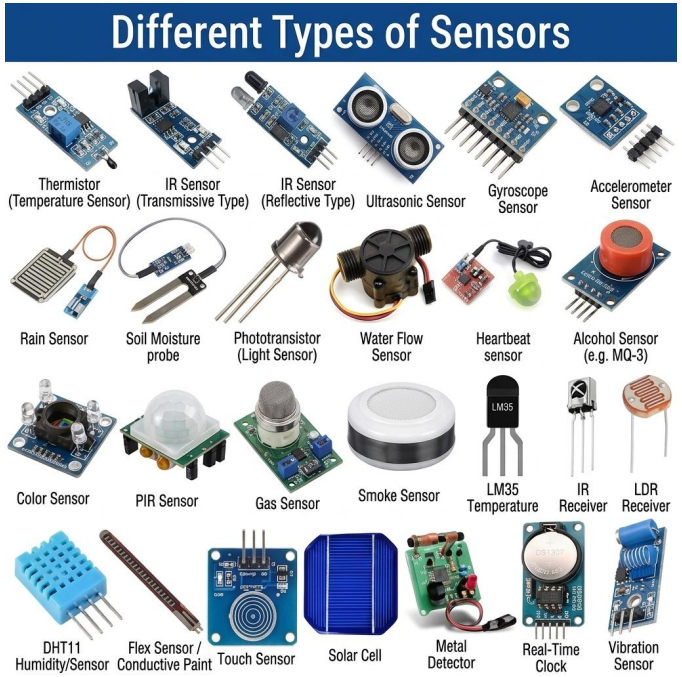
Microcontroller (Compute)

Resource-constrained processors with minimal RAM.

Actuators (Execution)

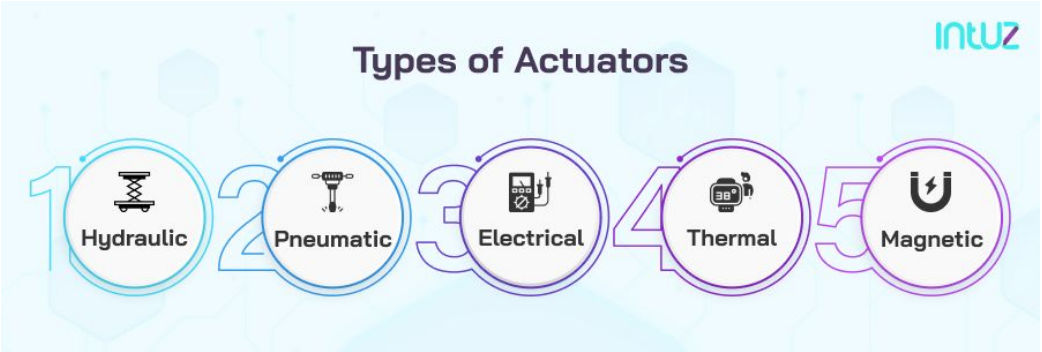
Electromechanical outputs altering physical reality.

Sensors and Actuators



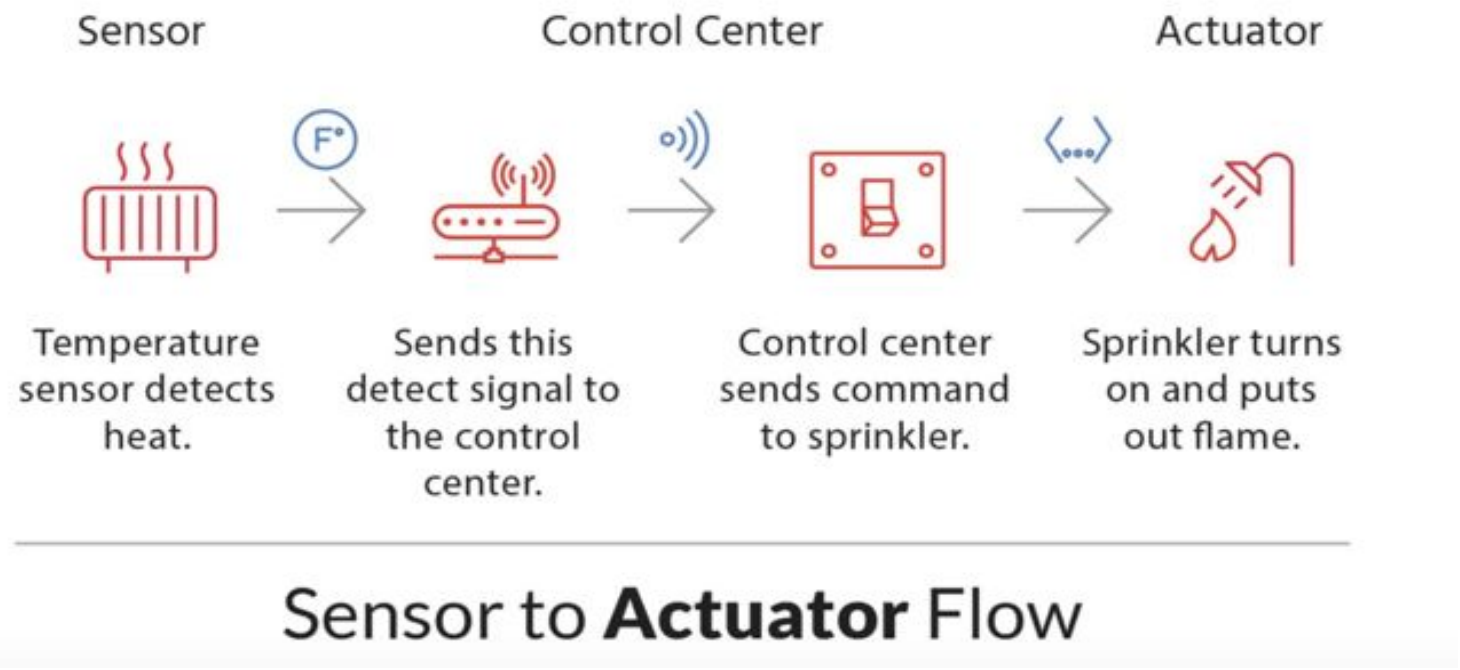
- **Sensors** or transducers represent physical devices that convert one form of energy into another.
- Sensors convert a physical device into an electrical impulse to take the desired action. For instance, sensors in an ambient light system will measure the brightness of the light by turning it into an electrical signal.

- **Actuators** convert electrical impulses into physical actions or objects.
- In the light example, as the sensor is reading the brightness of the light by converting it into an electrical signal, an actuator takes action according to the desired setting. It will decrease or increase the light brightness according to the set parameters.
- We can leverage actuators to control and manage our devices in the IoT network according to the information sent by the sensors.



Sensors and Actuators

Example of Sensor and Actuator



Traditional IoT vs. Edge (UbiComp) vs. Cloud-Centric IoT

Traditional IoT

Processing: Bare-metal execution on low-power microcontrollers.

Connectivity: Complete offline isolation with zero network dependency.

Scalability: Fixed rule-based logic without cloud telemetry or updates.

Edge Execution

Latency: Sub-millisecond deterministic reflex response.

Bandwidth: Massive uplink data volume reduction.

Autonomy: Uninterrupted operation during WAN outages.

Cloud Core IoT

Capacity: Elastic processing power and multi-petabyte scale.

Analytics: Fleet-wide artificial intelligence modeling.

Coordination: Cross-regional device registry management.

SYSTEM TOPOLOGY

Tiered Architecture

Traditional IoT

TIER 1
Perception
Sensors, transducers, and physical environmental sampling.

TIER 2
Communication
Protocol translation, packet batching, and local filtering.

TIER 3
Storage
Global time-series storage, model training, and business logic.

TIER 4
Application
End-user dashboards, alerts, and local control interfaces.

Cloud Core IoT

TIER 1
Perception
Sensors, transducers, and physical environmental sampling.

TIER 2
Network
Protocol translation, packet batching, and local filtering.

TIER 3
Middleware
Local cluster analytics and sub-second control loops.

TIER 4
Cloud Storage
Global time-series storage, model training, and business logic.

TIER 5
Application
Enterprise APIs, web/mobile apps, and user analytics.

Participatory vs. Opportunistic Sensing

Participatory Sensing

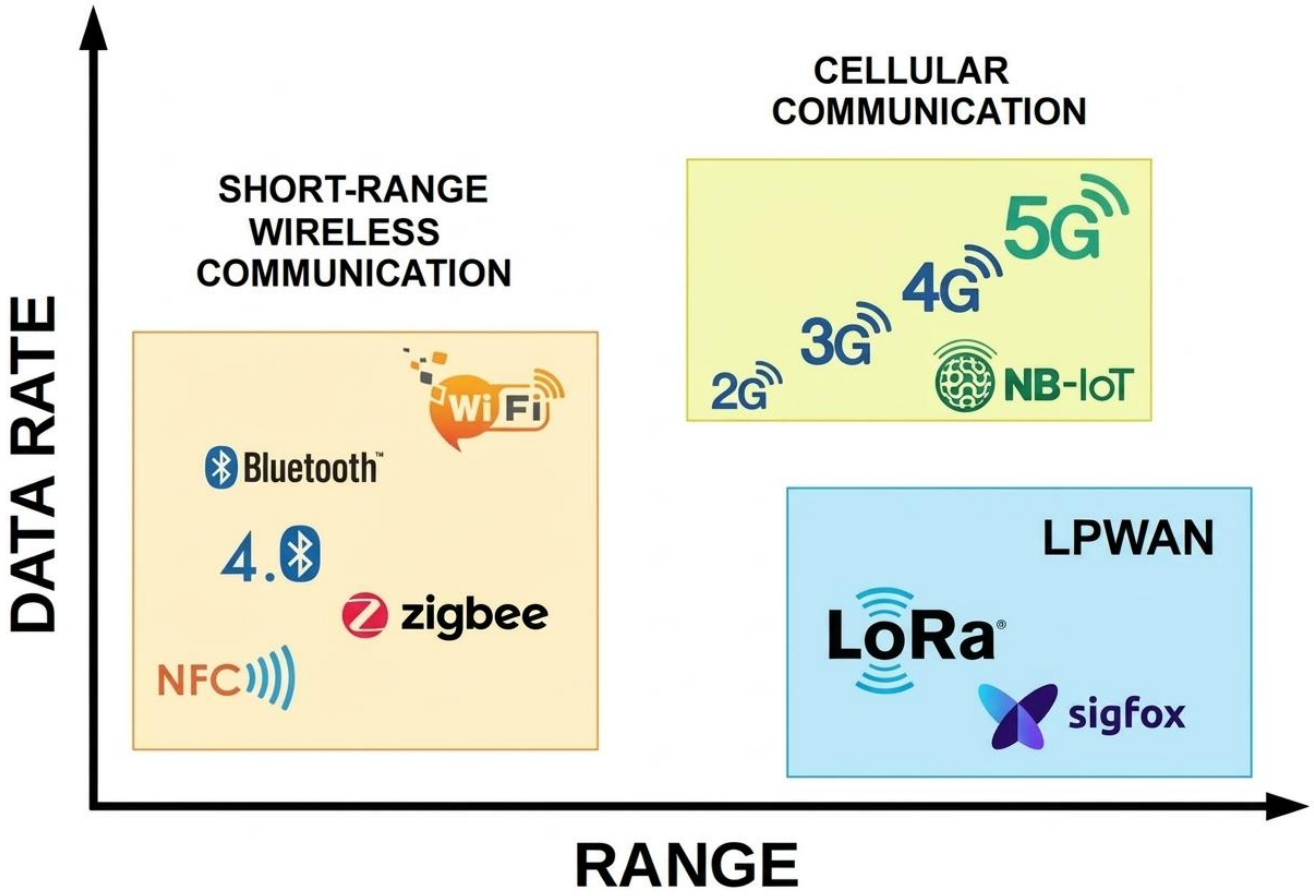
Users consciously record data like pothole photos and noise reports.

Opportunistic Sensing

Background vehicle telemetry and passive GPS congestion streams.



IoT Wireless Protocol Spectrum



CONNECTIVITY LAYER

IoT Wireless Protocol Standard

Standard	Typical Range	Data Throughput	Primary Use Case
Wi-Fi (802.11 n/ac/ax)	50 – 100 meters	54 Mbps – 9.6 Gbps	High-bandwidth video streaming, local gateways, and mains-powered smart home devices
Bluetooth (BLE 5.x)	10 – 100 meters	Up to 2 Mbps	Indoor beacons and wearable telemetry
Zigbee / 802.15.4	10 – 75 meters	250 Kbps	Home automation mesh networks
LoRaWAN	2 – 15 km	0.3 – 50 Kbps	Precision agriculture and utility meters
Cellular / NB-IoT / LTE-M	1 – 10 km	64 – 1000 Kbps	Licensed spectrum fleet tracking

Communication Protocol

HTTP

Client-Server Model (TCP)

Universal Support: Native web & cloud integration

Proven Security: Established TLS/HTTPS standard

Heavy Overhead: High header payload cost

Best: Cloud services & web dashboards

MQTT

Broker Pub/Sub Model (TCP)

Ultra Lightweight: Minimal payload overhead

Reliable Delivery: Configurable QoS tiers (0–2)

Broker Hub: Requires centralized infrastructure

Best: Telemetry, robotics & smart homes

CoAP

Constrained REST Model (UDP)

Lowest Power: Designed for battery micro-nodes

Multicast Native: One-to-many sensor queries

UDP Handling: Needs manual reliability logic

Best: Low-power wireless sensor networks

Technical Protocol Comparison

Dimension	HTTP	MQTT	CoAP
Transport Layer	TCP	TCP	UDP
Messaging Pattern	Request / Response	Publish / Subscribe	Request / Response + Multicast
Packet Overhead	**High (Large headers)**	**Very Low (2-byte header)**	**Ultra Low (4-byte header)**
Power Consumption	**High**	**Low**	**Minimal**
Infrastructure	Direct Web Server	Centralized Broker	Direct Gateway / Node
Network Spectrum	WiFi / Cellular	WiFi / Cellular	LoRaWAN / WiFi / Cellular

Selection Guideline: Use HTTP for cloud web tiers, MQTT for real-time telemetry pipelines, and CoAP for battery-constrained micro-controllers.

IoT Storage Paradigms

Time-Series (TSDB)

TimescaleDB, InfluxDB, QuestDB

Append-Only: Millions of inserts per second

Gorilla Encoding: 90% disk space reduction

Auto Downsampling: Continuous time aggregation

Document NoSQL

MongoDB, DynamoDB, Couchbase

Dynamic Schemas: Heterogeneous payload ingestion

Partition Sharding: Seamless horizontal scale-out

Native JSON: Direct payload ingestion without ETL

Relational (RDBMS)

PostgreSQL, MySQL

ACID Guarantee: Precise financial and asset records

Complex Joins: Link sensors with user metadata

Telemetry Tier

Raw sensor metrics stream

Real-time threshold alerts

Continuous downsampled rollups

Payload Archive Tier

Raw unparsed JSON payloads

Device error logs & crashes

Cold historical audit lakes

Metadata Tier

Device registries & auth

Customer site hierarchies

Billing & subscription tiers

STORAGE PARADIGMS

Technical Storage Comparison

Dimension	Time-Series (TSDB)	Document NoSQL	Relational (RDBMS)
Write Throughput	**Ultra-High (LSM/WAL)**	**High (Key-Sharded)**	**Moderate (Lock-Bound)**
Compression Ratio	**Exceptional (10×–20×)**	Moderate (Key Bloat)	**Low (Page-Level)**
Window Aggregations	**Sub-second Vectorized**	Moderate (Pipeline scan)	**Slow (Full Table Scan)**
Schema Elasticity	Semi-Flexible (Metrics)	**Fully Dynamic JSON**	**Rigid (DDL Migration)**
Device Asset Joins	**Limited**	Embedded References	**Superior (Full SQL)**

Trade-off Rule: Optimize TSDB for numerical stream depth; reserve RDBMS for relational breadth.

Semantic Intelligence

Ambiguous Telemetry

```
{"id": "node_04", "val": 24.5}
```

Missing physical unit, target feature, calibration date, and precision limits.

SOSA / SSN Standard

```
Observation -> AirTemperature (°C)
```

Explicit unit ontology, sensor provenance, and automated machine interoperability.

The Layer Transformation Pipeline Semantic

From Raw Sensor to Semantic Intelligence

1. Perception

RAW MEASUREMENT

`42°C`

Hardware outputs voltage or scalar floats without asset context.

2. Network

PACKET TRANSPORT

```
{"val": 42}
```

MQTT or CoAP delivers the uncontextualized binary or JSON payload.

3. Middleware

SEMANTIC ENRICHMENT

```
{"device":"MotorA", "state":"Overheating"}
```

Rules engines link sensor IDs with asset tables, thresholds, and operational meaning.

4. Application

BUSINESS ACTION

Ticket #MNT1001

ERP dispatches maintenance directly without human interpretation.

Semantic Web in IoT: Linked Data & SPARQL

RDF Triples

Representing entities via subject, predicate, and object triples.

JSON-LD Payloads

Embedding semantic URI context directly into lightweight JSON.

SPARQL Endpoints

Declarative cross-network graph queries querying physical systems.

Linked Data gives IoT measurements global context by converting raw key-value pairs into structured RDF graphs (`Subject -> Predicate -> Object`) using standards like **SOSA** and **JSON-LD**.

SPARQL lets developers search these graphs by defining relationship patterns rather than navigating fixed table joins.

Primary Applications: Smart buildings (combining HVAC, space management, and energy sensors), digital twins, complex supply chains, and cross-enterprise asset management.

EXERCISE CHALLENGE

System Design Scenario: Remote Wildfire Detection

Scenario Constraints

100 square kilometers of mountainous forest. Zero cellular coverage. Nodes operate on solar cells with battery backup.

Goal: Deliver alert to station in under 10 seconds.

Discussion Questions

1. Protocol: Would you select WiFi, Cellular or LoRaWAN? What protocol would you select? HTTP, MQTT, or CoAP?

2. Power: How will you duty-cycle sensors when smoke blocks sunlight?

3. Semantics: What metadata prevents false alarms across teams?

SUMMARY

Overall View

IoT architecture layer depth depends on scale, latency, and data complexity:

3-layer suits simple consumer apps transmitting raw sensor data;

4-layer adds an edge or middleware layer for local analytics and sub-second industrial control;

5-layer supports enterprise-scale ecosystems like smart cities requiring cloud storage, AI analytics, and business integration.

